AHMED OUARRALI

Full-Stack Software Engineer • Java / Spring Boot • Angular / React / Next.js • Seeking end-of-studies internship (PFE)

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EDUCATION

Engineering cycle in software engineering 2024 – present

ENSIASD, National School of Artificial Intelligence and Data Science, Taroudant

Artificial intelligence, data science, software architecture.

DUT in embedded computer engineering, with honors 2022 – 2024

EST Oujda, Higher School of Technology

Embedded systems, IoT, software development.

TECHNICAL SKILLS

Languages: Java 21, TypeScript, JavaScript, Python, PHP, Dart, C/C++ (Qt), SQL ; **Backend:** Spring Boot, Spring Security (JWT), Spring Data JPA, Hibernate, Laravel, Node.js, REST APIs, OpenAPI ; **Frontend:** Angular 21 (Signals, RxJS, Reactive Forms, Material), React, Next.js (App Router), Tailwind CSS, Flutter, internationalization ; **Databases:** PostgreSQL, MySQL, MongoDB, Redis, SQLite, Firebase, Flyway, modeling and indexing ; **DevOps & cloud:** Docker, Git, GitHub Actions, Google Cloud Run, Terraform, Prometheus, Grafana ; **Quality:** JUnit 5, Mockito, Testcontainers, Vitest, Testing Library, Playwright, Selenium WebDriver, Pytest, Cypress, ArchUnit, SonarQube, ESLint, JaCoCo, SpotBugs, axe-core ; **Embedded & IoT:** ESP32/ESP8266, Embedded C

PROFESSIONAL EXPERIENCE

Full-Stack Developer • Final-year project (PFA) Jun – Sep 2026

OdemLab, skincare platform: e-commerce, AI analysis, mobile app. Four-person team project

- Designed a **modular Spring Boot monolith** sliced vertically by business domain rather than technical layers, with **architecture tests** that fail the build when a module crosses a forbidden boundary.
- Implemented the **order pipeline**: pricing fully recomputed server-side (no client-supplied amount is ever used), ordered line locks against overselling, idempotency keys against duplicate orders, atomic reservation of coupons and loyalty points.
- Integrated **payment methods**: international card, Moroccan card with strong authentication, cash on delivery confirmed by one-time code, selected by market zone behind a single interface: adding an acquirer never touches the pipeline.
- **Reliability and delivery**: email outbox written in the business transaction then shipped under a distributed lock, circuit breaker per external service; deployment on **Google Cloud Run** with zero traffic, health checks, failover and **automatic rollback**.
- **Versioned schema** replayable from an empty database, verified foreign-key indexing; **API contract drift detected on every build**: any mismatch between server and frontend fails the build.

Technologies: Java 21 • Spring Boot • PostgreSQL • Redis • Next.js • React • TypeScript • Docker • Terraform • GitHub Actions

Full-Stack Developer • Internship Jul – Sep 2025

Medical appointment platform: matching patients with practitioners

- Built a healthcare app connecting patients and doctors: practitioner search, appointment booking and availability management.
- Designed role-based dashboards (patient, practitioner, admin) with graphical activity indicators.
- Modeled the database and implemented no-overlap rules for booking slots.

Technologies: PHP • Laravel • MySQL • JavaScript • AJAX • Bootstrap

Embedded Developer • Internship Mar – Jun 2024

MOUSSA SOFT, Agadir: cane-counting system with laser sensor

- Cane counting on a production line: VL53L0X laser sensor driven by ESP8266, OLED display and PHP/MySQL dashboard over Wi-Fi.

Technologies: ESP8266 • VL53L0X • PHP • MySQL

Test Automation Engineer • Internship Jun – Aug 2023

Univers Architecture Maroc, web application test automation

- Automated regression suites with Selenium WebDriver, replacing a manual campaign replayed on every release.
- Structured suites as reusable page objects, so a UI change ripples to a single point in the test code.
- Wrote the testing conventions later adopted by the team.

Technologies: Python • Selenium WebDriver • ChromeDriver

PROJECTS

ProcureFlow • Multi-tenant B2B procurement platform 2026

Requests, approvals, suppliers, orders, invoices and budgets, full-stack project

Modular Spring Boot monolith and Angular SPA: JWT authentication with refresh-token rotation, amounts recomputed server-side with idempotency keys, bilingual French/English interface, frontend types generated from the API contract, extraction history on MongoDB. Stepped approval workflow with bounded delegation, atomic budget reservation on approval, multi-tenant isolation where cross-tenant access answers 404, never 403. AI copilot: answers cited on workspace data, request drafts generated from natural-language descriptions.

Technologies: Java 21 • Spring Boot • Angular • TypeScript • PostgreSQL • MongoDB • Flyway • Playwright • Docker

Smart Campus Companion • Mobile app and augmented reality 2026

Campus navigation for ENSIASD Taroudant, full-stack mobile project

Indoor navigation in augmented reality (ARCore), interactive map and real-time room availability.

Technologies: Flutter • Dart • Firebase • Riverpod • ARCore • Node.js

FitTrack • Sports tracking platform 2025

Cloud-hosted web application, full-stack project

Training tracking with AI-assisted sessions, token authentication and containerized stack.

Technologies: Java 17 • Spring Boot • React • Next.js • TypeScript • Docker • MySQL

SmartSummarizer • Automated document processing 2025

Automatic summarization and quiz generation, machine learning project

Transformer-based summarization, quizzes and mind maps from course PDFs.

Technologies: Python • PyTorch • Hugging Face • spaCy • Flask

LANGUAGES

Arabic: native | French: professional | English: professional